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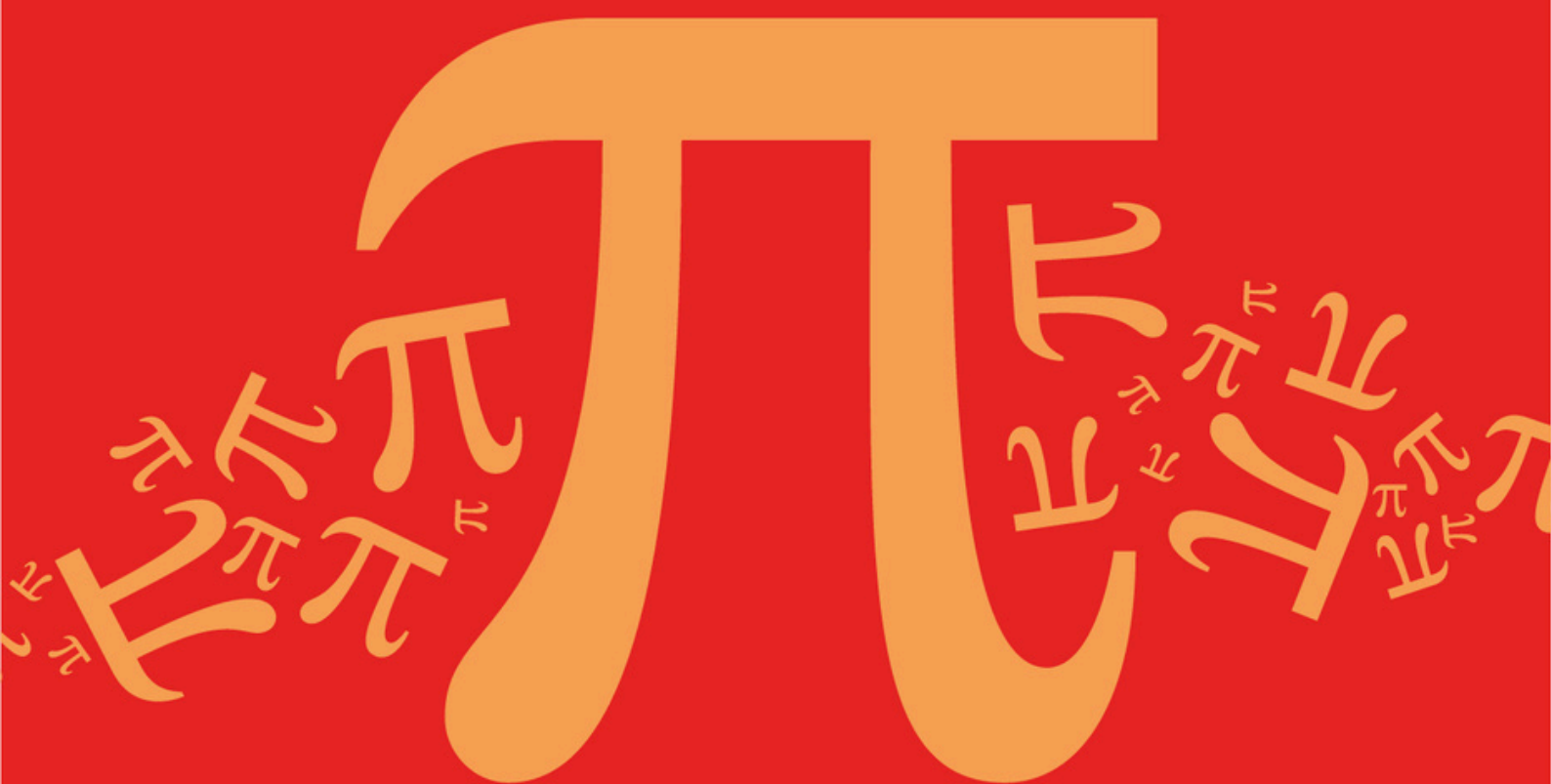
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Detailed mark scheme

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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C5: Determinants (2x2 Determinants)

Mark Scheme

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Mark Scheme

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C5: Determinants (2x2 Determinants)

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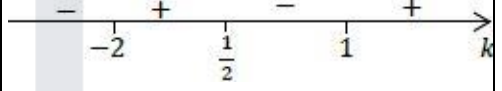
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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Substitutes $k = 1$ and correctly calculates the determinant, and concludes that the matrix is singular.	AO2.2a	B1	determinant = $4 \times 1 - 2 \times 2 = 0$ $\therefore$ the matrix is singular AG
(b)	Finds determinant in terms of k . Allow one error.	AO1.1a	M1	determinant = $k(5 - k) - 2(k^3 + 1)$
	Obtains a correct inequality in k .	AO1.1b	A1	$5k - k^2 - 2k^3 - 2 < 0$
	Obtains three correct critical values.	AO1.1b	A1	$2k^3 + k^2 - 5k + 2 > 0$ $(k - 1)(2k^2 + 3k - 2) > 0$ $(k - 1)(2k - 1)(k + 2) > 0$
	Deduces one correct region. FT their three real distinct critical values if given as $a < k < b$, $k > c$ (o.e.) where $a < b < c$	AO2.2a	A1F	
	Deduces the other correct region. FT their three real distinct critical values if given as $a < k < b$, $k > c$ (o.e.) where $a < b < c$. Condone the use of 'and'.	AO2.2a	A1F	
				$-2 < k < \frac{1}{2}, k > 1$
				Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds $k = 2$	AO1.1b	B1	$k - 2 = 0 \Rightarrow k = 2$
(b)	States correct transformation	AO1.2	B1	Reflection in the y-axis
(c) (i)	Finds product BA	AO1.1a	M1	$\mathbf{BA} = \begin{bmatrix} -1 & -2 \\ 1 & k \end{bmatrix}$
	Obtains inverse FT 'their' BA provided M1 awarded	AO1.1b	A1F	$(\mathbf{BA})^{-1} = \frac{1}{-k - (-2)} \begin{bmatrix} k & 2 \\ -1 & -1 \end{bmatrix}$
	Finds $\mathbf{A}^{-1}$ and $\mathbf{B}^{-1}$	AO1.1b	B1	$\mathbf{A}^{-1} = \frac{1}{k - 2} \begin{bmatrix} k & -2 \\ -1 & 1 \end{bmatrix}$

	Obtains correct $\mathbf{A}^{-1}$, $\mathbf{B}^{-1}$, and shows that $(k-2) \times (-1) = 2-k = -k - (-2)$ thus completing verification Must clearly show $\mathbf{A}^{-1} \times \mathbf{B}^{-1}$, method for this mark – disallow if answer simply stated	AO2.1	R1	$\mathbf{B}^{-1} = \frac{1}{-1} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\mathbf{A}^{-1}\mathbf{B}^{-1} = \frac{1}{(k-2) \times (-1)}$ $\begin{bmatrix} k+0 & (-2 \times -1)+0 \\ (-1 \times 1)+0 & 0+(-1 \times 1) \end{bmatrix}$ That is the same as $(\mathbf{BA})^{-1}$ $(\mathbf{BA})^{-1} = \frac{1}{2-k} \begin{bmatrix} k & 2 \\ -1 & -1 \end{bmatrix} = \mathbf{A}^{-1}\mathbf{B}^{-1}$
(ii)	Uses equation for identity from definition	AO3.1a	M1	We require to demonstrate that:
	Commences argument by manipulating the matrix products within the equation with clear pairing	AO2.1	R1	$(\mathbf{NM}) \times \{\mathbf{M}^{-1}\mathbf{N}^{-1}\} = \mathbf{I}$ $(\mathbf{NM}) \times \mathbf{M}^{-1}\mathbf{N}^{-1} = \mathbf{N}(\mathbf{M} \times \mathbf{M}^{-1})\mathbf{N}^{-1}$
	Clearly demonstrates that $\mathbf{M} \times \mathbf{M}^{-1} = \mathbf{I}$ used	AO2.4	B1	$= \mathbf{N} \mathbf{I} \mathbf{N}^{-1}$ $= \mathbf{N} \mathbf{N}^{-1}$
	Completes the argument using rigorous reasoning with definition of matrix inverse and associativity mentioned Must see all working with correct pairing of each matrix with inverse	AO2.1	R1	$= \mathbf{I}$ Using definition of matrix inverse and associativity of matrix multiplication Hence true for all non-singular matrices $\mathbf{N}$ and $\mathbf{M}$
Total 10 marks				

Q3.

- (a) Determinant of matrix = $-8 + 9 = 1$
Finding determinant and multiplying by area

M1

Area = $3 \times 1 = 3$ (square units)
CAO – must show multiplication or refer to scale factor / invariant area or equivalent

A1

2

(b) (i)
$$\begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x \\ mx+c \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$\Rightarrow$

$$(x') = 4x + 3(mx + c)$$

$$(y') = -3x - 2(mx + c)$$

x', y' in terms of x, y, m, c

M1

$$\text{Invariant lines} \Rightarrow y' = mx' + c$$

$$\Rightarrow -3x - 2mx - 2c = 4mx + 3m^2x + 3mc + c$$

$$\Rightarrow 0 = (3m^2 + 6m + 3)x + 3mc + 3c$$

Use of $y' = mx' + c$

A1

$$\Rightarrow 3m^2 + 6m + 3 = 0 \quad 3mc + 3c = 0$$

Attempt at solving equations where coefficients = 0
or compares coefficients

M1

$$3(m + 1)^2 = 0 \quad 3c(m + 1) = 0$$

$$\Rightarrow m = -1 \quad c \text{ can be any value}$$

Finding the correct value of m

A1

$$\Rightarrow \text{lines are } y = -x + c$$

Fully correct line – no restriction on c

A1

5

SPECIAL CASES

$$\begin{pmatrix} 4 & 3 \\ -3 & -2 \end{pmatrix} \begin{pmatrix} x \\ -x+c \end{pmatrix} = \begin{pmatrix} x+3c \\ -x-2c \end{pmatrix}$$

$$x' = x + 3c$$

$$y' = -x - 2c$$

SC1 – Correct multiplication as shown

Consider

$$\begin{aligned} & -x' + c \\ &= -(x + 3c) + c \\ &= -x - 3c + c \\ &= -x - 2c \\ &= y' \end{aligned}$$

Hence $y = -x + c$ is an invariant line

SC2 – correct multiplication as shown above and
full algebraic solution using $y' = -x' + c$

(ii) When $c = 0$, $y = -x$ is a line of invariant points

Any equivalent form

B1

1

[8]

Q4.

(a) Char. Eqn. is $\lambda^2 - 8\lambda - 9 = 0$

Attempted

M1

Quadratic solved to get two roots

dM1

$\Rightarrow \lambda = 9, -1$

A1

Substⁿ. back λ (at least once):

M1

$\lambda = 9 \Rightarrow -x + y = 0 \Rightarrow \lambda = 9$ has evecs. $\alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
any $\alpha \neq 0$

$\lambda = -1 \Rightarrow x + y = 0 \Rightarrow \lambda = -1$ has evecs. $\beta \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
any $\beta \neq 0$

A1

6

(b) (0, 0)

B1

1

[7]

Q5.

(a) $\mathbf{X} \mathbf{X}^T = \begin{bmatrix} 3 & x \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ x & 7 \end{bmatrix}$

Attempted multn. with X^T correct

M1

$$= \begin{bmatrix} x^2 + 9 & 7x - 3 \\ 7x - 3 & 50 \end{bmatrix}$$

A1

2

$$(b) \quad X^T X = \begin{bmatrix} 3 & -1 \\ x & 7 \end{bmatrix} \begin{bmatrix} 3 & x \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 10 & 3x - 7 \\ 3x - 7 & x^2 + 49 \end{bmatrix}$$

Good attempt

M1

$$X X^T - X^T X = \begin{bmatrix} x^2 - 1 & 4x + 4 \\ 4x + 4 & 1 - x^2 \end{bmatrix}$$

Good attempt

M1

$$\text{Det}(X X^T - X^T X) = \begin{vmatrix} x^2 - 1 & 4x + 4 \\ 4x + 4 & 1 - x^2 \end{vmatrix} = (x + 1)^2 \begin{vmatrix} x - 1 & 4 \\ 4 & 1 - x \end{vmatrix}$$

Good attempt to factorise / expand the determinant

M1

$$= -(x + 1)^2 \{(x - 1)^2 + 16\} \leq 0$$

for all real x

Explained / demonstrated fully

E1

4

$$(c) \quad x = -1$$

CSO

B1

1

[7]

Q6.

$$(a) \quad \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$$

B1

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

M1

$$= \begin{bmatrix} cX + sY \\ -sX + cY \end{bmatrix}$$

A1

3

(b) (i) $(cX + sY)^2 - 6(cX + sY)(-sX + cY) - 7(-sX + cY)^2 = 8$

$$(c^2X^2 + 2csXY + s^2Y^2) - 6([c^2 - s^2]XY) + sc\{Y^2 - X^2\} - 7(s^2X^2 - 2csXY + c^2Y^2) = 8$$

Substn. for x & y in eqn. and multiplying out

M1

$$p = c^2 + 6sc - 7s^2$$

A1

$$q = 16cs - 6(c^2 - s^2)$$

AG

A1

$$r = s^2 - 6sc - 7c^2$$

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(ii) Factorising:

$$3s^2 + 8sc - 3c^2 = (3s - c)(s + 3c) = 0$$

Or by double angles

M1A1

Deducing a tan value

M1

$$\tan \theta = \frac{1}{3} \quad (\theta \text{ acute})$$

A1

$$\cos \theta = \frac{3}{\sqrt{10}}, \sin \theta = \frac{1}{\sqrt{10}}$$

Both

A1

Substⁿ. sensible values back for p and r

M1

$$2X^2 - 8Y^2 = 8$$

A1

$$\frac{X^2}{2^2} - \frac{Y^2}{1^2} = 1$$

CSO

A1

8

- (iii) Since C' is a hyperbola, and it is just C rotated, it follows that C is a hyperbola

E1

1

[16]

Q7.

(a) $\text{Det}(\mathbf{M}) = -1$

B1

Magnitude = 1 $\Rightarrow$ area invariant
FT area s.f.

B1✓

–ve sign $\Rightarrow$ cyclic order of vertices is reversed
 OR “reflection” involved

B1

3

(b) **Method 1**

Char. Eqn.: $\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$

Finding and solving attempt

M1 A1

Substⁿ. back: $\lambda = 1 \Rightarrow y = \frac{1}{2}x$

M1 A1

and $\lambda = -1 \Rightarrow y = \frac{1}{4}x$

A1

5

Method 2

$$\begin{bmatrix} -3 & 8 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ mx \end{bmatrix} = \begin{bmatrix} (8m-3)x \\ (3m-1)x \end{bmatrix}$$

Attempted

(M1)

Use of $y' = mx'$: $3m - 1 = 8m^2 - 3m$

(M1)

Solving a quadratic eqn. in $m = \frac{1}{4}, \frac{1}{2}$
 From $(4m-1)(2m-1) = 0$

(M1A1)

$$p = \frac{1}{2} \text{ and } q = \frac{1}{4}$$

(A1)

(c) $p = \frac{1}{2} = \tan \theta$

$$\Rightarrow \cos 2\theta = \frac{3}{5} \text{ and } \sin 2\theta = \frac{4}{5}$$

For these attempted and used in a reflection matrix

©2025 Exam Papers Practice. All rights reserved. M1

$$\mathbf{R} = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix}$$

A1

2

[10]

Q8.

(a) $\det \mathbf{AB} = 110$

B1

Use of $\det \mathbf{AB} = \det \mathbf{A} \det \mathbf{B}$

M1

$\det \mathbf{B} = 11$

ft their det AB/ 10

A1F

3

(b) $\mathbf{C} = (\mathbf{AB})^T = \begin{bmatrix} 9 & 7 \\ 1 & 13 \end{bmatrix}$

M1A1

$$\mathbf{D} = [(\mathbf{BA})^T]^T = \mathbf{BA} = \begin{bmatrix} 14 & 2 \\ 1 & 8 \end{bmatrix}$$

For reference:

$$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 5 & 3 \\ -2 & 1 \end{bmatrix}$$

B1

3

[6]

Q9.

(a) $\begin{bmatrix} 1 & 4 & 2 \\ -1 & 2 & 6 \end{bmatrix} \begin{bmatrix} k & 1 \\ 2 & -1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} k+14 & -1 \\ 22-k & 3 \end{bmatrix}$

PQ a 2×2 matrix

M1

At least one element in C_1 correct

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All correct

A1

3

(b) $\text{Det}(\mathbf{PQ}) = 3k + 42 + 22 - k$

Det of a square matrix attempted and equated to zero

M1

$$= 2k + 64 = 0$$

$$k = -32$$

ft in 2×2 case only (linear eqns.)

A1

2

[5]

Q10.

(a) $\text{Det}(\mathbf{M}) = 1 \Rightarrow$ Area invariant under T

2nd B1 ft ref. "area"

B1B1

2

(b) Char. Eqn. $\lambda^2 - 2\lambda + 1 = 0$

M1

$\Rightarrow \lambda = 1$ (twice)

A1

Subst^g. their λ back to find an evect:

M1

$\alpha \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Any (non-zero) α

A1

(Since $\lambda = 1$) this represents a line of inv. pts.
ft if $\lambda \neq 1$

B1

5

(c) $\begin{bmatrix} -1 & 4 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ \frac{1}{2}x + k \end{bmatrix} = \begin{bmatrix} -x + 4k \\ -\frac{1}{2}x + 3k \end{bmatrix}$

M1A1

Verifying that $y' = \frac{1}{2}x' + k$
Be convinced AG

A1

3

[10]

Q11.

(a) For an *invariant line*, all points on the line have image points also on the line.

For a line of invariant points, all points on the line map onto themselves.

Clearly explained

B1

1

(b) Det = 3

B1

The s.f.. of area enlargement under T
Must mention area

B1
2

- (c) Setting $x' = x$ and $y' = y$ and solving
 $x = 2x - y$ and $y = -x + 2y$

M1

$$\Rightarrow y = x$$

A1

Or

Char. Eqn. is $\lambda^2 - 4\lambda + 3 = 0$

M1

$\lambda = 1$ gives l.o.i.p.s and $y = x$

A1
2

(d) (i)
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ -x + c \end{bmatrix}$$

M1

$$= \begin{bmatrix} 3x - c \\ -3x + 2c \end{bmatrix}$$

A1
2

- (ii) $y' = -x' + c$ also

M1

$$\Rightarrow y = -x + c \text{ invariant for all } c$$

A1
2

- (e) Stretch, s.f. 3,

M1A1

perp^r. to $y = x$ (or $\parallel$ to $y = -x$)

A1
3

Q12.

(a) $\begin{vmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{vmatrix} = 67.84 - 51.84 = 16$

Attempt at det.

M1A1

2

(b) Inv. pts.g.b. $x' = x, y' = y$

B1

Subst^g. in eqns. $6.4x - 7.2y = x$
 $-7.2x + 10.6y = y$

M2
A1

$$y = \frac{3}{4}x$$

A1

5

Alt. I

Char. Eqn. is $\lambda^2 - 17\lambda + 16 = 0$

M1A1

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$\lambda = 1$ or 16

A1

$\lambda = 1$ for l.o.i.p.s $\Rightarrow 5.4x - 7.2y = 0$

$$\text{i.e. } y = \frac{3}{4}x$$

ignore $\lambda = 16$ work

M1A1

(5)

Alt. II

$y = mx$ a l.o.i.p.s

$$\begin{bmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{bmatrix} \begin{bmatrix} x \\ mx \end{bmatrix} = \begin{bmatrix} 6.4x - 7.2mx \\ -7.2x + 10.6mx \end{bmatrix}$$

B1

$$-7.2x + 10.6mx = m(6.4x - 7.2mx) \text{ also}$$

M1
A1

$$\Rightarrow 7.2m^2 + 4.2m - 7.2 = 0$$

$$\Rightarrow (4m - 3)(3m + 4) = 0$$

A1

$$\Rightarrow y = \frac{3}{4}x \quad \text{or} \quad y = -\frac{4}{3}x$$

A1

Checking which one works

B1

(5)

[7]

Examiner reports

Q1.

The majority of students were able to calculate the determinant and show that it was equal to zero. However, a significant minority failed to make a conclusion after a correct calculation.

In part (b), most students were able to set up a correct inequality, although some wasted time searching for a root, having missed the connection with part (a). Those students who found the correct critical values, generally went on to identify the correct regions.

Q3.

This question proved to be the most demanding. Many students knew what to do for part (a) but often failed to show all working out leading to a loss of marks. Part (b)(i) proved elusive for all but the most able students. When asked to find all invariant lines the most efficient and valid method involves obtaining y' and x' in terms of x , y and c and then trying to make them fit the equation $y' = mx' + c$. This method was rarely seen and yet is the correct standard approach. Other approaches such as using eigenvalues are more appropriate to lines of invariant points only.

Q4.

This question was also generally well done. The only commonly lost mark was that for part (b) which should have been very straightforward; perhaps too straightforward.

Q5.

A significant number of candidates did not know what a transpose matrix is; 52 candidates failed to get the mark for writing down a correct transpose matrix for $\mathbf{X}$. Amongst those who did, it was a common mistake to attempt $\text{Det}(\mathbf{XX}^T - \mathbf{X}^T\mathbf{X})$ as $\text{Det}(\mathbf{XX}^T) - \text{Det}(\mathbf{X}^T\mathbf{X})$.

Even amongst those who attempted this build-up work with a correct method, algebraic errors were commonplace. More crucially, it was very disappointing indeed to see here (and in Question 8) that almost all candidates preferred to multiply out first, which meant that they were almost invariably stuck with a quartic expression that they stood very little chance of factorising. This led to the almost universal loss of at least two of the seven marks available on this question. Even those who tried to factorise out the obvious factor of $(x + 1)$ in each element of the matrix $\mathbf{XX}^T - \mathbf{X}^T\mathbf{X}$ failed to realise that they were “taking it out” of each row (or column) and so it was a repeated factor – without this, it was impossible to argue that the required determinant was always non-positive.

Of those candidates who obtained a correct quartic most did not distinguish between the words ‘negative’ and ‘non-positive’ and also stated that it must be negative because all the coefficients in the quartic were negative. Only a handful of candidates offered any explanation that $x^2 - 2x + 17 \geq 0$. Since the mark-scheme was designed to give the final mark in part (c) only to fully correct prior working, it was a mark lost by just about everyone; only 33 candidates scored more than 5 of the 7 marks available for the question.

Q6.

Although there were a few excellent, complete responses to this question, in general it

turned out to reflect the lack of preparedness on the part of candidates for the demands presented to them here. The three marks in part (a) were readily and easily acquired by most candidates attempting it. However, 156 candidates were unable to proceed beyond the given result of part (a), and made no attempt at the remaining parts of this question.

In part (b)(i), it was not uncommon to find that candidates were less than comfortable with squaring out their x and y terms in the new variables X and Y , and collecting them up proved to be far more problematical than should have been the case. Rather a lot of candidates seemed to think that the XY term came exclusively from the xy term, and such efforts foundered at this point.

Few candidates got to the correct equation of the hyperbola. A suitable answer to part (b)(iii) required only an awareness that the original curve had been rotated into the final hyperbola, but only a few candidates answered it without having worked their way to it. Of these, most offered spurious reasons, such as the not uncommon reference to the suggestion that the original curve was a hyperbola because this final one was an ellipse.

Q7.

In part (a), many candidates realised the significance of both the sign and magnitude of the determinant of the transformation matrix, although there are still far too many who try to describe area scale factors without using the key word “area”.

In part (b), almost all attempts managed to find or verify the given value of p , but most who did go on to try and find a value for q as well simply did what amounted to the same working again. The more successful bids found the eigenvalues first and then deduced p and q .

Attempts at part (c) fell almost equally into the three categories of correct, incorrect – usually a 45° or 90° rotation matrix – or nothing at all.

Q8.

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This question was intended to test a couple of basic areas of the module’s work on determinants and transposes. The former topic was generally handled very easily, but a majority of candidates appeared to resort to guessing what was involved in part (b) with the work on transposes. In general, candidates scored either 3 or 6 marks on this question.

Q9.

This was a straightforward starter to the paper, and was generally found to be so by candidates.

Almost all candidates knew what to do, in principle, although there were unexpectedly large numbers of arithmetical (particularly sign) errors in part (b) when finding the required value of k .

Q10.

This question required several different ideas. In part (a), it was essential to use the word “area” to describe the significance of **det M** in relation to **T**, although responses were much more frequently appropriate in this respect than had been the case the last time such a question appeared. A similar type of comment was required at the end of part (b). In part (c), those who parametrised the line as $(x, \frac{1}{2}x + k)$ generally coped very well, although a significant number lost the final mark by not showing the necessary working to

support the given answer carefully, as opposed to merely stating what had been stated in the question.

Q11.

In part (b), the determinant of the transformation matrix is the scale factor of area enlargement under T – very few candidates used the key word “area” here. In part (d), having found the image of $(x, -x + c)$ to be $(x', y') = (3x - c, 2c - 3x)$, which many successfully did, candidates were supposed simply to show that $y' = -x' + c$. Few candidates understood that this was what was required.

Traditionally, describing transformations is very poorly done, and this was no exception. There was a lack of understanding as to how the algebra, matrix algebra and coordinate geometry all tie together with the geometry of transformations, and some more practice was needed for most candidates on this score.

Q12.

Candidates found this question difficult. The answer to part (a) is the determinant of the given matrix; it was not uncommon for candidates to find this and then take its square root and offer this as the scale factor required. Although there are at least three valid approaches to part (b), most candidates' uncertainty led to a mish-mash of algebra, with the correct answer often embedded in so much other working that the candidate producing it seemed to have no idea that they had just done the necessary work.